

Joint Optimization of Path Planning and Cooperative Strategy for UAV–UGV Delivery

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Abstract—In this study, we consider a new cooperative unmanned aerial vehicle (UAV)–unmanned ground vehicle (UGV) delivery framework, where the UAV picks up parcels and the UGV drops off the parcels it is carrying to share the workload associated with picking up and delivering parcels. Because of the limited load capacity of the UAV, it can unload the parcels onto the UGV or at the destination. We mathematically formulate this system model and aim to optimize the horizontal and vertical trajectories of the two vehicles, as well as the binary indicators representing pickup, drop-off, and cooperation, to minimize the total mission completion time. To handle the nonconvexity of the formulated problem, we utilize a successive convex approximation technique to transform nonconvex constraints into convex sets. Additionally, we apply a penalty convex–concave procedure to relax the binary indicators to achieve continuous values for optimizations while preserving their binary characteristics. Finally, we propose a cooperative algorithm to iteratively derive suboptimal solutions from the relaxed convex problem. The simulation results demonstrate that the proposed scheme effectively optimizes both path planning and the cooperation strategy, enabling the UAV to drop off the parcels it carries onto the UGV at the optimal location. This approach significantly reduces delivery time and outperforms the baseline schemes in various environments.

Index Terms—Unmanned aerial vehicle, unmanned ground vehicle, path planning, cooperative strategy, convex optimization.

I. INTRODUCTION

THE demand for unmanned aerial vehicles (UAVs) has surged in recent years as innovations in miniaturization, battery technology, and sensor capabilities have significantly improved their performance and operational capabilities. In particular, the versatility, efficiency, and affordability of UAVs have revolutionized applications ranging from agriculture and environmental monitoring to disaster management and surveillance [1]. However, delivery service is one of the most prominent and rapidly expanding applications of UAVs. Major companies such as Amazon, UPS, and Uber are actively investing in UAV technology to deliver packages and food quickly and at a low cost over short distances. By integrating UAVs into their delivery operations, these

companies seek to mitigate challenges associated with urban traffic congestion, expensive labor, and lengthy delivery times, thereby enhancing the overall speed and efficiency of delivery services for consumers.

Despite the manifold benefits of UAV delivery, its operational capabilities face constraints primarily due to limited operational time and payload weight [2]. To increase the efficiency of UAV deliveries within these constraints, trajectory planning optimization plays an important role. For example, by determining the delivery sequence and optimal route to effectively transport items to the designated recipients, the UAV can minimize delivery time and energy consumption while maximizing the number of completed deliveries in a given amount of time. The abovementioned trajectory planning strategies are categorized into two main approaches according to the degree of involvement of the UAV.

The first approach is to use UAVs exclusively for short-distance delivery tasks, such as transporting medical supplies or delivering food between neighborhoods. To overcome the aforementioned fundamental limitations of UAVs, there is ongoing research on delivery strategies and trajectory designs that minimize delivery time [2], [3], [4], [5] or energy consumption [6], [7], [8]. In [2], two distinct UAV strategies were suggested: one aimed at minimizing costs while adhering to delivery time constraints, and another focused on reducing delivery times within a limited budget. Similarly, UAV flight routes were optimized to expedite parcel deliveries while ensuring consistent coverage across designated neighborhoods in [3]. Addressing dynamic task scheduling and minimizing tardiness in on-demand meal deliveries, UAV operations were optimized under varying task arrival times and deadlines in [4]. In addition, the joint optimization of multiple trajectories and pickup strategy of the UAV was conducted to minimize delivery time while ensuring pickup of all parcels in [5]. Regarding energy efficiency, the integration of UAVs into electric vehicle routing problems was explored to minimize energy consumption under maximum payload constraints in [6]. Furthermore, UAV speeds and launch point selections were optimized to reduce energy usage within constrained service times and ranges [7]. Additionally, UAV trajectory planning strategies were investigated to minimize energy consumption and handoff frequencies, taking into account limitations such as battery capacity and connectivity issues in [8].

The second approach entails integrating UAVs with unmanned ground vehicles (UGVs), leveraging their combined capabilities to address the challenges posed by the restricted

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payload capacity and operational duration of UAVs [9], [10], [11], [12], [13], [14], [15]. A framework for computer vision-aided cooperative UAV-UGV path planning was proposed in [9]. In particular, the UAV captured images of the environment around the UGV and used computer vision algorithms to identify obstacles for the UGV to avoid. In [10], the respective circular path plans for UAVs and UGVs were designed based on a simple sweep-based approach and an online local adjustment strategy to minimize their travel times while maintaining complete coverage. In [11], a UAV-UGV collaboration based on the robot operating system was developed to create a low-cost aerial-ground delivery system for residential neighborhoods. Further, in [12], a dynamic aggregation strategy-based path planner was introduced to determine the most time-efficient route for a capability-limited UGV and a fuel-constrained UAV to cover all mission points. Specifically, Voronoi diagrams and greedy set cover algorithms were used to determine the number of meeting points needed to cover all work points on the road network. The A-star and Lin-Kernighan heuristic algorithms were also used to plan distance-optimal routes for UGVs along the road network. Finally, the hybrid adaptive large neighborhood search algorithm was applied to design time-optimal routes for UAVs. Further, cooperative UGV-UAV nested vehicle routing planning was solved by a mixed integer linear program, which can be applied to emergency resource delivery [13]. Furthermore, an upper-lower hierarchical deep learning-based control framework was proposed in [14], which includes a gated recurrent unit recurrent deep neural network-based motion planning algorithm and a deep reinforcement learning-based online collision avoidance method. The recent research on the UAV-UGV cooperation was well summarized in terms of methodologies and applications in [15].

Cooperation between UAVs and UGVs has great importance and potential for realistic delivery services, as they can compensate for each other's shortcomings. This cooperation also reduces the distance UAVs fly compared to a solo delivery, which can lower safety risks in the workplace. However, most of the existing research has focused on designing individual path plans rather than selecting the time and location for cooperation considering the optimal routes for both vehicles. Accordingly, we explore practical UAV and UGV cooperative division of labor delivery scenarios that have not yet been addressed. In particular, the UGV transports items from the warehouse to various delivery zones, while the UAV collects items from pickup zones and can drop off parcels onto either the UGV or at the final destination, ensuring that the parcel weight does not exceed the maximum load during transit. Further, we determine the best time and location to drop off the items picked up by the UAV onto the UGV, which is the main difference from the previous study. The contributions of our study can be summarized in two aspects.

i) We provide a mathematical model of a realistic UAV-UGV cooperative delivery that allows the UAV to unload the collected parcels onto the UGV or the destination to overcome the limited load capacity. In particular, the mathematical modeling of the two vehicles meeting at the same place at the same time with time synchronization to exchange

parcels is a novel idea that has not been addressed in previous research.

ii) To achieve this, we aim to jointly optimize the horizontal and vertical trajectories of the two vehicles and binary indicators for pickup, drop-off, and cooperation to minimize the total mission completion time, but the problem is formulated as mixed-integer nonlinear programming (MINLP). To address the nonconvex nature of the problem under consideration, we propose advanced optimization methods that have not been exploited in UAV-UGV cooperation so far. We initially employ the successive convex approximation (SCA) technique to transform the problem into a convex form and utilize the penalty convex-concave procedure (PCCP) to maintain the binary nature of the relaxed integer variables during optimization. Based on these relaxations, we propose a cooperative algorithm that applies interior-point methods to the transformed convex problem to iteratively find the suboptimal strategy, and provide an analysis of the computational complexity. Through comprehensive simulations across various scenarios, we confirm that the proposed scheme can effectively design path planning and a cooperation strategy to drop off the parcels carried by the UAV onto the UGV at the optimal location. This scheme also surpasses the baseline schemes in reducing mission completion time under different system parameters.

The remainder of this paper is organized as follows. In Section II, we present the system model of cooperative UAV-UGV delivery along with problem formulation. In Section III, we propose a UAV-UGV cooperative algorithm using SCA, PCCP, and interior-point methods. In Section IV, we assess the performance of the proposed scheme by comparing it with baseline schemes under different system parameters. Finally, we provide the concluding remarks in Section V.

II. SYSTEM MODEL

We consider cooperative UAV-UGV delivery, where $m \in \mathcal{M} = \{G, U\}$ is an indicator of the type of vehicle, e.g., U for UAV and G for UGV. In particular, the UGV has K parcels, each assigned to a specific destination, while the UAV collects L parcels from designated pickup zones, where the set of parcels to be delivered is denoted as $\mathcal{K} = \{1, 2, \dots, K\}$, and the set of parcels to be picked up is denoted as $\mathcal{L} = \{1, 2, \dots, L\}$. More specifically, both vehicles depart from a starting point, the UGV drops all the parcels it is carrying in each delivery zone, the UAV picks up all parcels from each pickup zone, and they finally arrive at a destination. To achieve this, the UGV and UAV must approach delivery zones and pickup zones within tolerances, respectively. Moreover, each vehicle has a maximum load capacity. Therefore, the UAV can drop off parcels onto the UGV (*case 1*) or at the destination (*case 2*) to ensure that the weight of the parcels in transit does not exceed the maximum load capacity, as shown in Fig. 1. Specifically, for case 1, the two vehicles must meet at the same time and place in order for the UAV to unload the parcels it is carrying on the UGV. For stable cooperation between the two vehicles, it is essential to consider constraints that ensure reliable communication between the UAV and

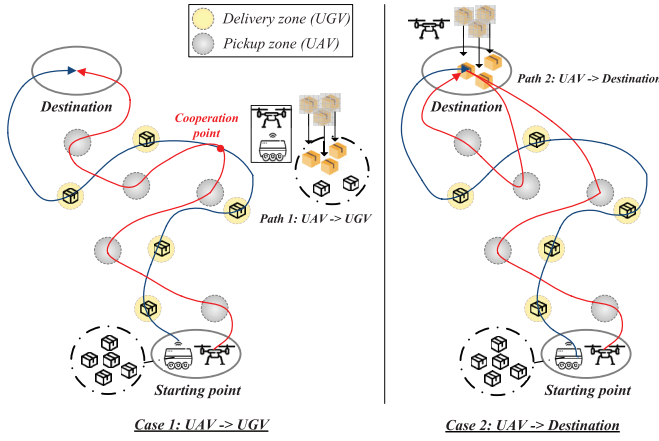


Fig. 1. System model of cooperative UAV-UGV delivery.

UGV. In addition, causal constraints related to the pickup and drop-off of parcels must be considered, as the UAV can drop off the parcel either to the UGV or at the destination after picking it up. The mission is completed when the UGV delivers all K parcels and the UAV picks up all L parcels and both vehicles arrive at the destination. The goal is then to optimize the cooperation strategy between them to minimize the mission completion time.

The scenario described above can be mathematically formulated as follows. We first assume that there are N time slots, and the 3D coordinates of vehicle m at time slot $n \in \mathcal{N} = \{1, 2, \dots, N\}$ are represented by $\mathbf{c}_m[n] = (\mathbf{q}_m[n], z_m[n])$, where $\mathbf{q}_m[n] = (x_m[n], y_m[n])$ and $z_m[n]$ are horizontal and vertical coordinates, respectively. It is also assumed that the UAV flies at a permissible flight altitude, such as $H_{\min} \leq z_U[n] \leq H_{\max}$, whereas the UGV moves on the ground with $z_G[n] = 0$. With the predetermined starting point and destination, the related constraints for the two vehicles are given by

$$\mathbf{c}_m[0] = \mathbf{c}_{\text{str}}, \quad \forall m, \quad (1)$$

$$\mathbf{q}_U[1] = \mathbf{q}_{\text{str}}, \quad (2)$$

$$\mathbf{c}_m[N] = \mathbf{c}_{\text{dst}}, \quad \forall m, \quad (3)$$

$$\mathbf{q}_U[N-1] = \mathbf{q}_{\text{dst}}, \quad (4)$$

$$\mathbf{q}_m[n] \in [\mathbf{q}_{\min}, \mathbf{q}_{\max}], \quad \forall m, n, \quad (5)$$

where $\mathbf{c}_{\text{str}} = (\mathbf{q}_{\text{str}}, z_{\text{str}})$ and $\mathbf{c}_{\text{dst}} = (\mathbf{q}_{\text{dst}}, z_{\text{dst}})$ are the coordinates of the starting point and destination, respectively. Notably, constraints (2) and (4) prevent the UAV from leaving the allowed flight altitude by completing takeoff and landing within the starting point and destination, respectively. Furthermore, constraint (5) represents the boundary constraint on the area in which the two vehicles can move [16], ensuring that the positions of both vehicles remain within the defined boundary for all time slots.

The parcel $i \in \mathcal{K} \cup \mathcal{L}$ has a weight w_i and a vehicle m has a maximum deliverable load of W_m ; therefore, it cannot carry all items simultaneously when the total weight of parcels in transit at time slot n , denoted by $w_m[n]$, exceeds its maximum load, i.e., $w_m[n] \geq W_m$. Given that $W_G \gg W_U$ generally holds [17], the UAV drops the parcels it picked up by time

slot n onto the UGV or at the destination to avoid being unable to pick up any more parcels owing to weight restrictions. To design the mathematical model for the drop-off and pickup of parcels, let $\alpha_k[n]$ denote a drop-off indicator for parcel $k \in \mathcal{K}$ by the UGV, where $\alpha_k[n] = 1$ indicates that parcel k is dropped off at a delivery zone by the UGV at time slot n , and $\alpha_k[n] = 0$ otherwise. Similarly, let $s_l^+[n]$ and $s_l^-[n]$ denote the pickup and drop-off indicators for parcel $l \in \mathcal{L}$ by the UAV at time slot n , respectively. We also define $\beta_1[n]$ and $\beta_2[n]$ as indicators of whether the UAV unloads the parcels it is carrying onto the UGV or at the destination, respectively. For example, $\beta_1[n] = 1$ represents that the parcels carried by the UAV are dropped off at the UGV at time slot n , and $\beta_1[n] = 0$ otherwise. Because $\beta_1[n]$ and $\beta_2[n]$ cannot be 1 simultaneously, we define $\beta[n] = \beta_1[n] + \beta_2[n]$ to indicate whether the UAV unloaded its loads on the way. In other words, $s_l^+[n]$ and $s_l^-[n]$ are indexes representing the state of each parcel while $\beta_1[n]$ and $\beta_2[n]$ are indexes representing the behavior of the UAV, where the UAV decides whether to drop off the collected parcels at the UGV or at the destination. By defining $\{s_l^+[n], s_l^-[n], \beta_1[n], \beta_2[n]\}$, we can use the values of $\beta_1[n]$ and $\beta_2[n]$ to determine where the UAV drops the parcels halfway, and the values of $s_l^+[n]$ and $s_l^-[n]$ to express the parcels currently being carried and the parcels to be dropped. Therefore, we can formulate the following constraints:

$$\alpha_k[n] \in \{0, 1\}, \quad \forall k, n, \quad (6)$$

$$s_l^+[n], s_l^-[n] \in \{0, 1\}, \quad \forall l, n, \quad (7)$$

$$\beta_1[n], \beta_2[n] \in \{0, 1\}, \quad \forall n, \quad (8)$$

$$\sum_{n=1}^N \alpha_k[n] \leq 1, \quad \forall k, \quad (9)$$

$$\sum_{n=1}^N s_l^+[n] \leq 1, \quad \forall l, \quad (10)$$

$$\sum_{j=2}^n s_l^-[j] \leq \sum_{j=1}^{n-1} s_l^+[j], \quad \forall l, n \neq 1, \quad (11)$$

$$s_l^+[n] + \beta_1[n] + \beta_2[n] \leq 1, \quad \forall l, n, \quad (12)$$

$$\sum_{l=1}^L \sum_{j=1}^{n-1} (s_l^+[j] - s_l^-[j]) \geq \beta[n], \quad \forall n \neq 1, \quad (13)$$

$$s_l^-[n] \leq \beta[n], \quad \forall l, n, \quad (14)$$

$$1 - \left(\sum_{j=1}^n (s_l^+[j] - s_l^-[j]) \right) \geq \beta[n], \quad \forall l, n, \quad (15)$$

$$w_U[n] = \sum_{l=1}^L w_l \left(\sum_{j=1}^n (s_l^+[j] - s_l^-[j]) \right) \leq W_U, \quad \forall n, \quad (16)$$

$$w_G[n] = \sum_{k=1}^K w_k \left(1 - \sum_{j=1}^n \alpha_k[j] \right) + \sum_{l=1}^L w_l \left(\sum_{j=1}^n s_l^-[j] \right) \beta_1[n] \leq W_G, \quad \forall n. \quad (17)$$

Constraint (9) indicates that parcel k must be delivered by the UGV only once, and constraint (10) implies that parcel

l must be picked up by the UAV only once during the entire time slot. Constraint (11) represents the condition that the UAV cannot drop off each parcel before picking it up and constraint (12) means that the UAV cannot pick up and drop off the parcel simultaneously. Moreover, constraint (13) denotes the condition that the UAV can only unload parcels if there are parcels being carried, constraint (14) indicates that no parcels can be dropped until the UAV decides to drop off parcels onto the UGV or at the destination, and constraint (15) implies that all parcels being carried must be dropped off if the UAV decides to drop off parcels onto the UGV or at the destination. Finally, (16) and (17) are constraints on the maximum load for each vehicle.

To drop off or pick up parcels, the UGV (UAV) needs access to delivery zones with a tolerance of $\mathcal{R}_k$ (pickup zones with a tolerance of $\mathcal{R}_l$), which can be formulated as follows:

$$\alpha_k[n] \|\mathbf{q}_G[n] - \mathbf{q}_k\|^2 \leq \mathcal{R}_k^2, \quad \forall k, n \neq N, \quad (18)$$

$$s_l^+[n] (\|\mathbf{q}_U[n] - \mathbf{q}_l\|^2 + z_U^2[n]) \leq \mathcal{R}_l^2, \quad \forall l, n \neq N, \quad (19)$$

$$s_l^+[n] \|\mathbf{q}_U[j] - \mathbf{q}_l\|^2 \leq \mathcal{R}_l^2, \quad \forall l, n \neq N, j \in \{n-1, n+1\}, \quad (20)$$

where $\mathbf{q}_k = (x_k, y_k)$ and $\mathbf{q}_l = (x_l, y_l)$ are the horizontal center coordinates of the k -th delivery zone and l -th pickup zone, respectively, and delivery and pickup zones are assumed to be located on the ground for simplicity, such as $z_k = z_l = 0$. It is important to note that parcel k must be delivered to its designated delivery zone $\mathbf{q}_k$ and parcel l must be picked up from its designated pickup zone $\mathbf{q}_l$. In addition, when each vehicle approaches a delivery or pickup zone within the tolerance, $\alpha_k[n]$ and $s_l^+[n]$ can be set to 1. Constraints (19) and (20) indicate that the UAV must be within this pickup zone both when retrieving the parcel and while descending to pick up the parcel and ascending after picking up the parcel to guarantee the allowed flight altitude.

Similarly, to drop off all the parcels it picks up, the UAV must approach the UGV or destination within a tolerance of $\mathcal{R}_H$. Therefore, we need the following constraints:

$$\beta_1[n] (\|\mathbf{q}_U[n] - \mathbf{q}_G[n]\|^2 + z_U^2[n]) \leq \mathcal{R}_H^2, \quad \forall n \neq N, \quad (21)$$

$$\beta_1[n] \|\mathbf{q}_U[j] - \mathbf{q}_G[j]\|^2 \leq \mathcal{R}_H^2, \quad \forall n \neq N, j \in \{n-1, n+1\}, \quad (22)$$

$$\beta_2[n] (\|\mathbf{q}_U[n] - \mathbf{q}_{dst}\|^2 + z_U^2[n]) \leq \mathcal{R}_H^2, \quad \forall n \neq N, \quad (23)$$

$$\beta_2[n] \|\mathbf{q}_U[j] - \mathbf{q}_{dst}\|^2 \leq \mathcal{R}_H^2, \quad \forall n \neq N, j \in \{n-1, n+1\}. \quad (24)$$

Furthermore, the UAV can only descend from the permitted altitude to the ground when picking up or dropping off the parcel, which can be formulated as

$$\left(1 - \sum_{l=1}^L s_l^+[n] - \beta[n]\right) H_{\min} \leq z_U[n] \leq H_{\max}, \quad \forall n. \quad (25)$$

For cooperation, it is essential to ensure reliable communication between the UAV and UGV. To this end, we define

the channel power gain between the two vehicles, assuming line-of-sight propagation [18], as

$$h[n] = \frac{v_0}{d[n]^2} = \frac{v_0}{\|\mathbf{c}_U[n] - \mathbf{c}_G[n]\|^2}, \quad \forall n, \quad (26)$$

where $d[n]$ is the distance between the two vehicles at time slot n and v_0 is the average channel power gain at a reference distance of 1 m. Subsequently, the achievable spectral efficiency is given by

$$r[n] = \log_2 \left(1 + \frac{h[n] p_{tx}}{\sigma^2}\right) = \log_2 \left(1 + \frac{v_0 p_{tx}}{\sigma^2 \|\mathbf{c}_U[n] - \mathbf{c}_G[n]\|^2}\right), \quad (27)$$

where p_{tx} is the constant transmit power used by the two vehicles to communicate and σ^2 is the noise power. Finally, the communication requirement for cooperation can be formulated as follows:

$$r[n] \geq r_{\min}, \quad \forall n, \quad (28)$$

where $r_{\min}$ is the minimum spectral efficiency required for cooperation. With some manipulation, constraint (28) can be transformed to the following convex set.

$$\frac{v_0 p_{tx}}{(2^{r_{\min}} - 1) \sigma^2} \geq \|\mathbf{c}_U[n] - \mathbf{c}_G[n]\|^2, \quad \forall n. \quad (29)$$

To complete the pickup and drop-off of all parcels, the following conditions must be satisfied:

$$\sum_{k=1}^K \sum_{n=1}^N \alpha_k[n] \geq K, \quad (30)$$

$$\sum_{l=1}^L \sum_{n=1}^N s_l^+[n] \geq L. \quad (31)$$

The total distance traveled by vehicle m to complete a delivery can be expressed as

$$D_m = \sum_{n=1}^N d_m[n] = \sum_{n=1}^N \|\mathbf{c}_m[n] - \mathbf{c}_m[n-1]\|. \quad (32)$$

If vehicle m is assumed to travel at its maximum velocity V_m in every time slot,¹ the mission completion time of vehicle m can be calculated as

$$T_m = \frac{D_m}{V_m}. \quad (33)$$

To reflect the delay caused by unexpected traffic [19], we adopt a stationary spatio-temporal Poisson process [20], where the probability mass function for given time T_0 and distance D_0 is defined as follows:

$$P(N(D_0, T_0) = x) = \frac{\Lambda(D_0, T_0)^x e^{-\Lambda(D_0, T_0)}}{x!}. \quad (34)$$

¹Notably, this assumption is reasonable as each vehicle must move at its maximum velocity in every time slot to minimize delivery time.

Here, the cumulative intensity $\Lambda(D_0, T_0)$ is given by

$$\Lambda(D_0, T_0) = \oint_{D_0} \int_{T_0} \lambda_0 dt dx dy = \lambda_0 D_0 T_0, \quad (35)$$

where λ_0 is the rate of traffic occurrences per unit time and unit length.

Then, the expectation of the cumulative intensity is

$$\mathbb{E}[\Lambda(D_0, T_0)] = \sum_{x=0}^{\infty} x \cdot \left(\frac{\lambda_0 D_0 T_0 e^{-\lambda_0 D_0 T_0}}{x!} \right) = \lambda_0 D_0 T_0. \quad (36)$$

Using (36), the mission completion time of the UGV can be modified as

$$T'_G = \frac{D_G}{V_G} + \lambda_0 D_G T_G \mu, \quad (37)$$

where μ is the delay caused by unit traffic generation.

We aim to optimize the drop-off indicator of the UGV ($\mathbf{A} \triangleq \{\alpha_k[n], \forall k, n\}$), the pickup, drop-off, and cooperation indicators of the UAV ($\mathbf{S} \triangleq \{s_l^+[n], s_l^-[n], \forall l, n\}$ and $\mathbf{B} \triangleq \{\beta_1[n], \beta_2[n], \forall n\}$), and the trajectories of both vehicles ($\mathbf{Q} \triangleq \{\mathbf{q}_m[n], \forall m, n\}$ and $\mathbf{Z} \triangleq \{z_U[n], \forall n\}$) to minimize the total mission completion time. Accordingly, the optimization problem can be formulated as follows:

$$\begin{aligned} (\mathbf{P0}): \quad & \min_{\mathbf{A}, \mathbf{S}, \mathbf{B}, \mathbf{Q}, \mathbf{Z}} \quad \eta = T_U + T'_G \\ & \text{s. t.} \quad (1) - (25), (29) - (31). \end{aligned}$$

Problem (P0) is a nonconvex MINLP because $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$ are the sets of binary variables, e.g., constraints (6)–(8), and constraints (17)–(24) are not convex sets with respect to (w.r.t.) the relevant optimization variables. It is therefore challenging to derive a globally optimal solution mathematically.

III. PROPOSED COOPERATIVE ALGORITHM

To deal with the binary variables of $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$, we relax them to have continuous values and use PCCP to maintain their binary nature. We also apply SCA to transform nonconvex constraints into convex sets, allowing traditional convex optimization solvers such as CVX [21] to find solutions based on interior-point methods. Finally, we propose an iterative algorithm to solve the relaxed convex problem. The specific procedure to solve each nonconvex constraint is detailed below.

We use the following equivalent form to handle the constraints for binary variables, e.g., (6)–(8):

$$0 \leq \tau \leq 1, \quad (38)$$

$$\tau(1 - \tau) \leq 0, \quad (39)$$

where $\tau \in \{\alpha_k[n], s_l^+[n], s_l^-[n], \beta_1[n], \beta_2[n], \forall k, l, n\}$. Note that constraint (39) ensures that the binary nature of τ is preserved; however, it is still difficult to solve because it is not a convex set and the search space for optimizing τ is narrow. Using the fact that the left-hand side of (39) is concave w.r.t. τ and that a concave function is upper-bounded by its first-order Taylor expansion at any point [22], we can derive the upper bound of $\tau(1 - \tau)$, and (39) can be modified as follows:

$$(\tau^r)^2 + \tau(1 - 2\tau^r) \leq \rho, \quad (40)$$

where τ^r is the value of τ updated for the r -th iteration and $\rho \in \boldsymbol{\rho} = \{\rho_k^\alpha[n], \rho_{l,+}^s[n], \rho_{l,-}^s[n], \rho_1^\beta[n], \rho_2^\beta[n], \forall k, l, n\}$ is a nonnegative slack variable used for PCCP. Notably, the initial feasible region of efficient updates to optimize τ can be increased by replacing 0 with ρ in (40).

To handle the nonconvexity of (17), we replace $\left(\sum_{j=1}^n s_l^-[j]\right)\beta_1[n]$ with its equivalent form, as follows:

$$\begin{aligned} & \left(\sum_{j=1}^n s_l^-[j]\right)\beta_1[n] \\ &= \frac{\left(\beta_1[n] + \sum_{j=1}^n s_l^-[j]\right)^2 - \left(\beta_1^2[n] + \left(\sum_{j=1}^n s_l^-[j]\right)^2\right)}{2}. \end{aligned} \quad (41)$$

In (41), the terms including $\beta_1^2[n]$ and $\left(\sum_{j=1}^n s_l^-[j]\right)^2$ are convex and a convex function is lower-bounded by its first-order Taylor expansion at any point [22]; therefore, we can derive their respective lower bounds as follows:

$$\beta_1^{\text{LB}}[n] \triangleq 2\beta_1^r[n]\left(\beta_1[n] - \beta_1^r[n]\right) + \left(\beta_1^r[n]\right)^2, \quad (42)$$

$$\begin{aligned} \left(\sum_{j=1}^n s_l^-[j]\right)^{\text{LB}} &\triangleq 2\left(\sum_{j=1}^n s_l^{-r}[j]\right)\left(\sum_{j=1}^n s_l^-[j] - \sum_{j=1}^n s_l^{-r}[j]\right) \\ &\quad + \left(\sum_{j=1}^n s_l^{-r}[j]\right)^2, \end{aligned} \quad (43)$$

where $\beta_1^r[n]$ and $s_l^{-r}[j]$ are the values of $\beta_1[n]$ and $s_l^-[j]$ updated for the r -th iteration, respectively.

Using (42) and (43), we can also derive the upper bound of (41) as follows:

$$\begin{aligned} & \left(\sum_{j=1}^n s_l^-[j]\right)\beta_1[n] \\ &\leq \frac{\left(\beta_1[n] + \sum_{j=1}^n s_l^-[j]\right)^2 - \left(\beta_1^{\text{LB}}[n] + \left(\sum_{j=1}^n s_l^-[j]\right)^{\text{LB}}\right)}{2} \\ &\triangleq \zeta_l^{\text{UB}}[n], \quad \forall n, l. \end{aligned} \quad (44)$$

By substituting $\zeta_l^{\text{UB}}[n]$ into constraint (17), it can be replaced with the following convex set:

$$w_G[n] \leq \sum_{k=1}^K w_k \left(1 - \sum_{j=1}^n \alpha_k[j]\right) + \sum_{l=1}^L w_l \zeta_l^{\text{UB}}[n] \leq W_G, \quad \forall n. \quad (45)$$

To address the nonconvexity of (18), we transform it into the following equivalent form:

$$\alpha_k[n] \leq \frac{\mathcal{R}_k^2 + \varepsilon^2}{\|\mathbf{q}_G[n] - \mathbf{q}_k\|^2 + \varepsilon^2}, \quad (46)$$

where ε is a positive constant that prevents division by zero. Constraint (46) is still not a convex set, but its RHS is convex

w.r.t. $\|\mathbf{q}_G[n] - \mathbf{q}_k\|^2 + \varepsilon^2$. Therefore, we can derive its lower bound using the first-order Taylor expansion as follows:

$$\begin{aligned} & \frac{\mathcal{R}_k^2 + \varepsilon^2}{\|\mathbf{q}_G[n] - \mathbf{q}_k\|^2 + \varepsilon^2} \\ & \geq \frac{\mathcal{R}_k^2 + \varepsilon^2}{\|\mathbf{q}_G^r[n] - \mathbf{q}_k\|^2 + \varepsilon^2} \\ & \quad - \frac{\mathcal{R}_k^2 + \varepsilon^2}{(\|\mathbf{q}_G^r[n] - \mathbf{q}_k\|^2 + \varepsilon^2)^2} \left(\|\mathbf{q}_G[n] - \mathbf{q}_k\|^2 - \|\mathbf{q}_G^r[n] - \mathbf{q}_k\|^2 \right) \\ & \triangleq \gamma_k^\alpha[n]. \end{aligned} \quad (47)$$

where $\mathbf{q}_G^r[n]$ is the value of $\mathbf{q}_G[n]$ updated for the r -th iteration.

Using (47), we can replace constraint (18) with the following convex set:

$$\alpha_k[n] \leq \gamma_k^\alpha[n], \quad \forall k, n \neq N. \quad (48)$$

Using a similar method, we can convert constraints (19)–(24) into convex sets as follows:

$$s_l^+[n] \leq \gamma_l^s[n], \quad \forall l, n \neq N, \quad (49)$$

$$s_l^+[n] \leq \gamma_l^s[j], \quad \forall l, n \neq N, j \in \{n-1, n+1\}, \quad (50)$$

$$\beta_1[n] \leq \gamma_1^\beta[n], \quad \forall n \neq N, \quad (51)$$

$$\beta_1[n] \leq \gamma_1^\beta[j], \quad \forall n \neq N, j \in \{n-1, n+1\}, \quad (52)$$

$$\beta_2[n] \leq \gamma_2^\beta[n], \quad \forall n \neq N, \quad (53)$$

$$\beta_2[n] \leq \gamma_2^\beta[j], \quad \forall n \neq N, j \in \{n-1, n+1\}, \quad (54)$$

where $\gamma_l^s[n]$, $\gamma_l^s[j]$, $\gamma_1^\beta[n]$, $\gamma_1^\beta[j]$, $\gamma_2^\beta[n]$, and $\gamma_2^\beta[j]$ are given by

$$\begin{aligned} & \gamma_l^s[n] \\ & \triangleq \frac{\mathcal{R}_l^2 + \varepsilon^2}{\|\mathbf{q}_U^r[n] - \mathbf{q}_l\|^2 + (z_U^r[n])^2 + \varepsilon^2} \\ & \quad - \frac{\mathcal{R}_l^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[n] - \mathbf{q}_l\|^2 + (z_U^r[n])^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[n] - \mathbf{q}_l\|^2 + z_U^2[n] - \|\mathbf{q}_U^r[n] - \mathbf{q}_l\|^2 - (z_U^r[n])^2 \right), \end{aligned} \quad (55)$$

$$\begin{aligned} & \gamma_l^s[j] \\ & \triangleq \frac{\mathcal{R}_l^2 + \varepsilon^2}{\|\mathbf{q}_U^r[j] - \mathbf{q}_l\|^2 + \varepsilon^2} - \frac{\mathcal{R}_l^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[j] - \mathbf{q}_l\|^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[j] - \mathbf{q}_l\|^2 - \|\mathbf{q}_U^r[j] - \mathbf{q}_l\|^2 \right), \end{aligned} \quad (56)$$

$$\begin{aligned} & \gamma_1^\beta[n] \\ & \triangleq \frac{\mathcal{R}_H^2 + \varepsilon^2}{\|\mathbf{q}_U^r[n] - \mathbf{q}_G^r[n]\|^2 + (z_U^r[n])^2 + \varepsilon^2} \\ & \quad - \frac{\mathcal{R}_H^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[n] - \mathbf{q}_G^r[n]\|^2 + (z_U^r[n])^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[n] - \mathbf{q}_G[n]\|^2 + z_U^2[n] - \|\mathbf{q}_U^r[n] - \mathbf{q}_G^r[n]\|^2 - (z_U^r[n])^2 \right), \end{aligned} \quad (57)$$

$$\begin{aligned} & \gamma_1^\beta[j] \\ & \triangleq \frac{\mathcal{R}_H^2 + \varepsilon^2}{\|\mathbf{q}_U^r[j] - \mathbf{q}_G^r[j]\|^2 + \varepsilon^2} - \frac{\mathcal{R}_H^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[j] - \mathbf{q}_G^r[j]\|^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[j] - \mathbf{q}_G[j]\|^2 - \|\mathbf{q}_U^r[j] - \mathbf{q}_G^r[j]\|^2 \right), \end{aligned} \quad (58)$$

$$\begin{aligned} & \gamma_2^\beta[n] \\ & \triangleq \frac{\mathcal{R}_H^2 + \varepsilon^2}{\|\mathbf{q}_U^r[n] - \mathbf{q}_{dst}\|^2 + (z_U^r[n])^2 + \varepsilon^2} \\ & \quad - \frac{\mathcal{R}_H^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[n] - \mathbf{q}_{dst}\|^2 + (z_U^r[n])^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[n] - \mathbf{q}_{dst}\|^2 + z_U^2[n] - \|\mathbf{q}_U^r[n] - \mathbf{q}_{dst}\|^2 - (z_U^r[n])^2 \right), \end{aligned} \quad (59)$$

$$\begin{aligned} & \gamma_2^\beta[j] \\ & \triangleq \frac{\mathcal{R}_H^2 + \varepsilon^2}{\|\mathbf{q}_U^r[j] - \mathbf{q}_{dst}\|^2 + \varepsilon^2} - \frac{\mathcal{R}_H^2 + \varepsilon^2}{(\|\mathbf{q}_U^r[j] - \mathbf{q}_{dst}\|^2 + \varepsilon^2)^2} \\ & \quad \times \left(\|\mathbf{q}_U[j] - \mathbf{q}_{dst}\|^2 - \|\mathbf{q}_U^r[j] - \mathbf{q}_{dst}\|^2 \right), \end{aligned} \quad (60)$$

where $\mathbf{q}_U^r[n]$ and $z_U^r[n]$ are the values of $\mathbf{q}_U[n]$ and $z_U[n]$ updated for the r -th iteration, respectively.

With the modified convex constraints, problem (P0) can be transformed into the following convex optimization problem:

$$\begin{aligned} \text{(P1):} \quad & \min_{\mathbf{A}, \mathbf{S}, \mathbf{B}, \mathbf{Q}, \mathbf{Z}, \boldsymbol{\rho}} T_U + T_G' + \kappa P(\boldsymbol{\rho}) \\ & \text{s. t.} \quad (1) - (5), (9) - (16), (25), (29) - (31), \\ & \quad (38), (40), (45), (48) - (54), \end{aligned}$$

where $P(\boldsymbol{\rho}) = \sum_{k=1}^K \sum_{n=1}^N \rho_k^\alpha[n] + \sum_{n=1}^N (\rho_1^\beta[n] + \rho_2^\beta[n]) + \sum_{l=1}^L \sum_{n=1}^N (\rho_{l,+}^s[n] + \rho_{l,-}^s[n])$. Note that $P(\boldsymbol{\rho})$ is the penalty term that forces $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$ to be binary and $\kappa > 0$ is a variable that controls the effect of the penalty term on the objective function of (P1). For example, if the value of κ is small, the variables are optimized to minimize the total mission completion time $T_U + T_G'$ while allowing for some flexibility so that $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$ do not strictly adhere to their binary property. However, as κ increases, the variables are optimized to not only minimize $T_U + T_G'$ but also force $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$ to have binary values to make the penalty term zero [23].

Algorithm 1 summarizes the overall procedure of the proposed cooperative algorithm to find the solution from the relaxed convex problem iteratively.

Remark 1 (Convergence): Algorithm 1 starts from a feasible initial point $\{\mathbf{A}^0, \mathbf{S}^0, \mathbf{B}^0, \mathbf{Q}^0, \mathbf{Z}^0\}$ and a penalty control variable κ^0 , and increases κ^0 by a constant $\sigma > 1$ at each iteration until it approaches the limit bound $\kappa_{\max}$. According to Theorem 1 of [24], there exists a finite value of $\kappa_{\max}$, which guarantees $P(\boldsymbol{\rho}) \rightarrow 0$. From the point where $P(\boldsymbol{\rho}) = 0$, due to the iterative updating of the solution in a decreasing direction, the optimal solution of (P1) at the r -th iteration, denoted as $\{\mathbf{A}^r, \mathbf{S}^r, \mathbf{B}^r, \mathbf{Q}^r, \mathbf{Z}^r\}$, satisfies the following inequality for given

Algorithm 1 Proposed Cooperative Algorithm

```

1: Set  $r=0$ 
2: Initialize  $\mathbf{A}^r, \mathbf{S}^r, \mathbf{B}^r, \mathbf{Q}^r, \mathbf{Z}^r, \kappa^r, \kappa_{\max}, \sigma > 1$ 
3: Initialize the objective function of (P1), denoted by  $f^r$ ,
   to any
   arbitrary value
4: repeat
5:    $r \leftarrow r + 1$ 
6:    $f^{\text{old}} \leftarrow f^{r-1}$ 
7:   For given  $\{\mathbf{A}^{r-1}, \mathbf{S}^{r-1}, \mathbf{B}^{r-1}, \mathbf{Q}^{r-1}, \mathbf{Z}^{r-1}\}$ , find
      $\{\mathbf{A}^r, \mathbf{S}^r, \mathbf{B}^r, \mathbf{Q}^r, \mathbf{Z}^r\}$  by solving (P1) and evaluating  $\rho$ 
8:   Update  $\kappa^r = \min\{\sigma \kappa^{r-1}, \kappa_{\max}\}$ 
9:   Calculate  $f^r = T_U^r + T_G^r + \kappa^r P(\rho)$ 
10: until  $|f^r - f^{\text{old}}| \leq \epsilon$ 

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the previous value $\{\mathbf{A}^{r-1}, \mathbf{S}^{r-1}, \mathbf{B}^{r-1}, \mathbf{Q}^{r-1}, \mathbf{Z}^{r-1}\}$.

$$f(\mathbf{A}^r, \mathbf{S}^r, \mathbf{B}^r, \mathbf{Q}^r, \mathbf{Z}^r) \leq f(\mathbf{A}^{r-1}, \mathbf{S}^{r-1}, \mathbf{B}^{r-1}, \mathbf{Q}^{r-1}, \mathbf{Z}^{r-1}). \quad (61)$$

This implies that the objective value of problem (P1) is nonincreasing with each iteration and is lower-bounded by a finite value [25], thereby ensuring the convergence of the proposed algorithm.

Remark 2 (Computational Complexity): Adopting the methodology for assessing the computational complexity of the worst-case interior point method [22], [26], the proposed algorithm has a computational complexity of $O((\max(M, K, L)N)^{3.5} R_C \log(1/\epsilon))$, where R_C is the number of iterations required to converge the loop (lines 4–10). Therefore, the proposed algorithm exhibits a polynomial time computational complexity of $\max(M, K, L)N$ with an exponent of 3.5, making it highly suitable for real-time operations [27].

IV. RESULTS AND DISCUSSIONS

For performance evaluation, we consider the following parameters as default values [5], [28], [29], [30], [31], [32], [33]: $M = 2$, $K = 5$, $L = 4$, $N = 60$, $H_{\min} = H_{\max} = 10$ [m], $V_G = 8$ [m/s], $V_U = 15$ [m/s], $W_G = 60$ [kg], $W_U = 10$ [kg], $\mathcal{R}_H = 1$ [m], $\mathcal{R}_k = \mathcal{R}_l = 3$ [m], $r_{\min} = 0.53$ [bps/Hz], $v_0 = 0$ [dB], $p_{\text{tx}} = 30$ [dBm], $\sigma^2 = 104$ [dBm], $\mu = 31$ [s/traffic], $\lambda_0 = 1.28 \times 10^{-6}$ [traffic/m·s], $\kappa^0 = 15$, $\kappa_{\max} = 10^6$, $\sigma = 1.1$, and $\epsilon = 10^{-3}$. Moreover, the delivery and pickup zones were randomly determined within a rectangular area of size 800 m $\times$ 800 m, i.e., $\mathbf{q}_{\min} = (-400, -400)$ m and $\mathbf{q}_{\max} = (400, 400)$ m, and the value of ϵ was determined by trial and error for a given deployment.

For performance comparison, we consider the following four schemes:

- 1) *Proposed scheme*: The path planning and cooperative strategy of the UAV and UGV is determined by Algorithm 1.
- 2) *Non-Cooperation scheme*: The path planning of the UAV and UGV is determined by Algorithm 1, but they do not cooperate, such that the UAV can drop off parcels at the destination but not onto the UGV.

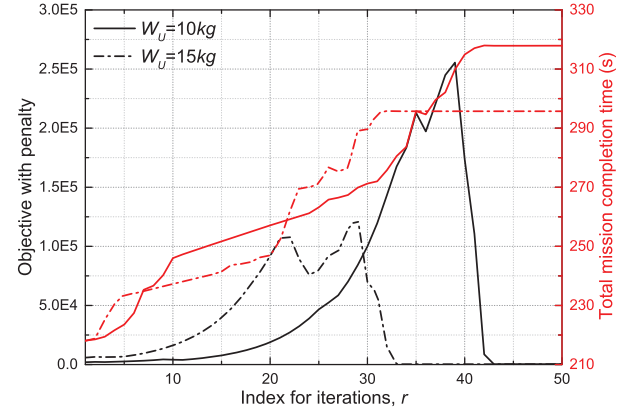


Fig. 2. Convergence behavior of the proposed scheme.

- 3) *Lightweight-First scheme*: The path planning and cooperative strategy of the UAV and UGV is determined to prioritize the drop-off and pickup of lightweight parcels.
- 4) *Heavyweight-First scheme*: The path planning and cooperative strategy of the UAV and UGV is determined to prioritize the drop-off and pickup of heavyweight parcels.
- 5) *RRT* scheme* [34]: The pickup and drop-off order and cooperation locations are determined the same as in the proposed scheme, but the path planning is optimized by a rapidly exploring random tree star (RRT*) algorithm with 2×10^4 iterations.

Fig. 2 shows the convergence behavior of the proposed scheme under different maximum loads for the UAV (W_U). The objective with the penalty in (P1) initially rises rapidly because the penalty function becomes more important as κ increases and ρ remains nonzero. During this period, the relaxed binary variables $\mathbf{A}$, $\mathbf{S}$, and $\mathbf{B}$ are optimized across a wider feasible region owing to the nonzero value of ρ , allowing them to converge to their optimal values with each iteration. With sufficient iterations, these variables attain binary values, leading to the penalty term approaching zero. Consequently, the objective with the penalty declines and eventually converges to a stationary point. Moreover, the proposed scheme shows a much higher objective with the penalty when $W_U = 10$ kg than when $W_U = 15$ kg. This is because the smaller W_U is, the more the UAV has to cooperate with the UGV, and therefore the larger the number of parameters in ρ used in the penalty function. However, the total mission completion time (η) grows monotonically and finally converges as the variables are optimized with each iteration. Furthermore, η converges to a lower value faster when $W_U = 15$ kg than when $W_U = 10$ kg, indicating that a larger load capacity of the UAV reduces the number of additional collaborations, resulting in faster delivery.

Fig. 3 shows the trajectories and resource allocations with the proposed and non-cooperative schemes: subfigures (a)–(c) for the proposed scheme when $W_U = 10$ kg, subfigures (d)–(f)

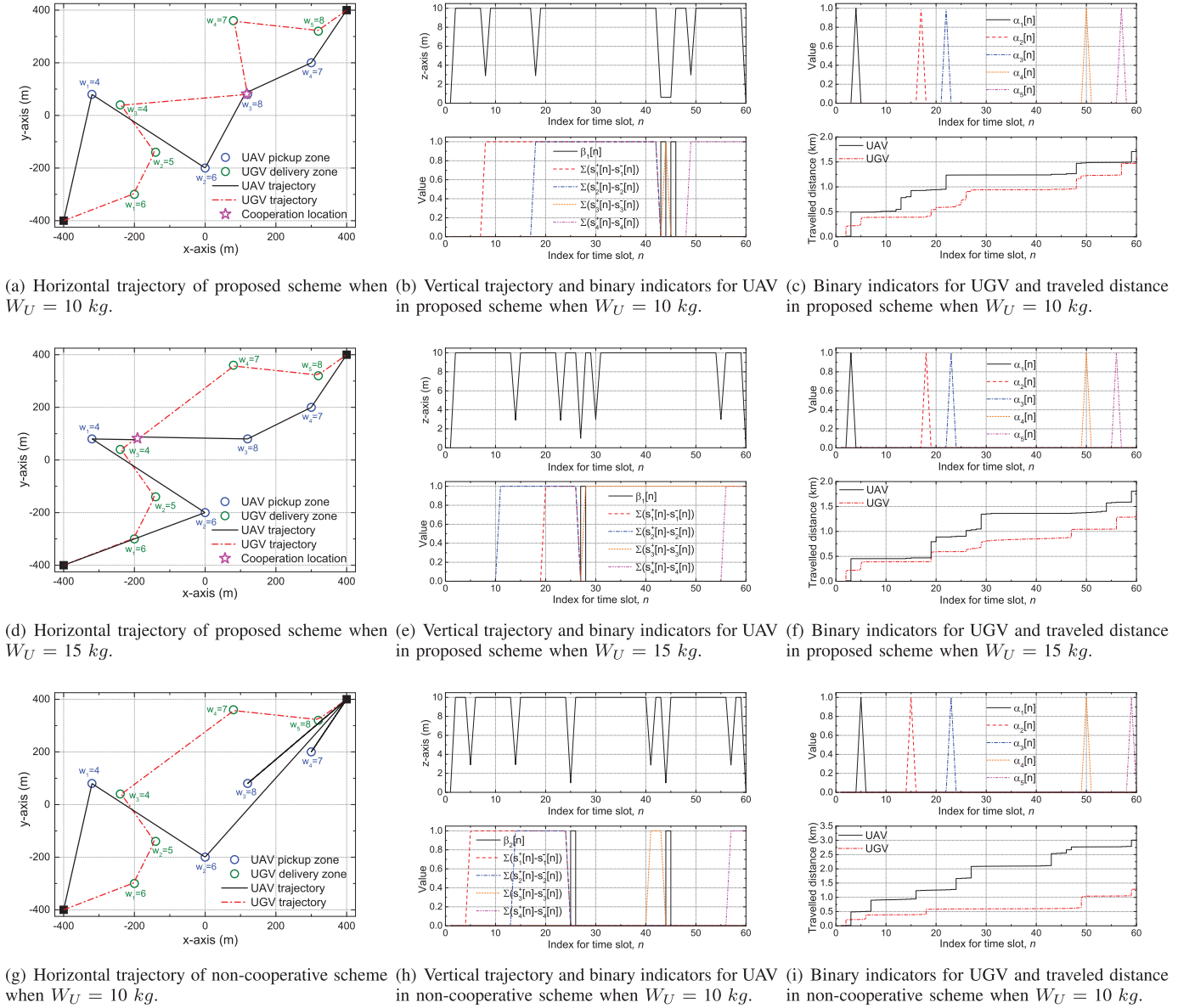


Fig. 3. Trajectories and resource allocations with the proposed and non-cooperative schemes.

for the proposed scheme when $W_U = 15$ kg, and subfigures (g)–(i) for the non-cooperative scheme when $W_U = 10$ kg.²

In Fig. 3(a), the pickup and delivery zones are depicted by blue and green circles, respectively, and the weight of the parcel in each zone is annotated. This result confirms that the UGV visits delivery zones to drop off the parcels it is carrying and the UAV visits pickup zones to pick up the parcels. The UAV reaches its maximum allowable weight when picking up the first and second parcels; therefore, it meets the UGV at the third pickup zone and drops off these parcels. Interestingly, after the UAV drops off the first and second parcels it is already carrying onto the UGV, it immediately picks up the third parcel and drops it back onto the UGV. This means that at this point,

²Because nonconvex problems have many local optima and the convergence point depends on the initial values of the optimization variables [22], [23], we use multiple random initializations and run the algorithm several times to select the best solution.

the UAV performs two collaborations. We can observe this phenomenon in detail in Fig. 3(b). The UAV flies at an altitude of 10 m and lands at each pickup zone within a tolerance of 3 m only when picking up parcels and within a tolerance of 1 m when dropping off parcels onto the UGV. We also see that the binary variables related to the UAV are well set to be either 0 or 1. When cooperating with the UGV, the UAV drops all the parcels it is carrying, and $\beta_1[n]$ is set to 1 a total of two times around the third pickup zone. Fig. 3(c) shows that the drop-off indicator of the UGV is also set to binary owing to the PCCP method and the distances traveled by the UAV and UGV increase as they perform their delivery duties.

As shown in Figs. 3(d)–3(f), as the maximum load for the UAV increases to 15 kg, it can carry more parcels at a time. Therefore, it collaborates with the UGV only once while traveling from the first pickup zone to the third pickup zone. Accordingly, the UGV forms an efficient path

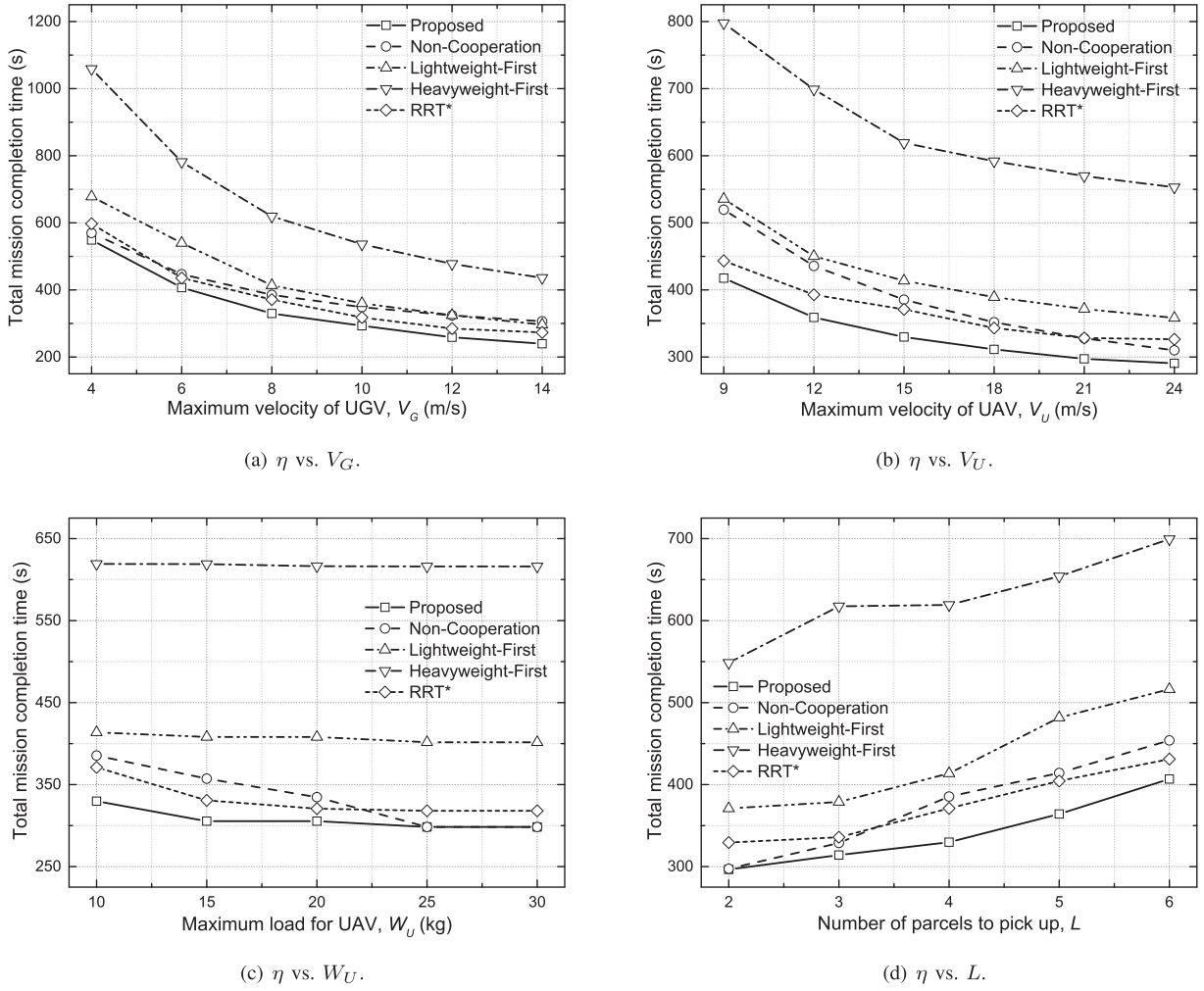


Fig. 4. Performance comparisons with respect to different system parameters.

for single cooperation, which is different from the path in Fig. 3(a). Figs. 3(g)–3(i) show the results for the non-cooperative scheme. In this scheme, the UAV cannot cooperate with the UGV; therefore, if the parcels it is carrying exceed the maximum allowable weight, it must drop off the parcels at the destination. Consequently, the UAV forms an inefficient path, and the traveled distance significantly increases as shown in Fig. 3(i). Furthermore, Fig. 3(h) shows that when the UAV drops off parcels at the destination, instead of $\beta_1[n]$, $\beta_2[n]$ is set to 1 and all the parcels carried by it are dropped off. We also confirm from these results that the vertical trajectory of the UAV and binary variables work well with their respective constraints.

Fig. 4 shows the total mission completion time (η) versus different system parameters: (a) maximum velocity of the UGV (V_G), (b) maximum velocity of the UAV (V_U), (c) maximum load for the UAV (W_U), and (d) number of parcels to pick up (L).

Figs. 4(a) and 4(b) show that the η of all schemes improves as V_G and V_U increase. This is because both vehicles can complete their delivery missions faster owing to the enhanced maneuverability. The lightweight-first and heavyweight-first schemes can cooperate with the UGV but perform worse

than the non-cooperation scheme because the fixed parcel scheduling order makes it more likely that they will form inefficient routes depending on the configuration of pickup and delivery zones. By contrast, the proposed scheme achieves the shortest η compared to the baseline schemes over the entire range of V_G and V_U . Fig. 4(c) shows the impact of the maximum load for the UAV on performance. As W_U increases, the UAV can complete its mission without cooperation with the UGV, allowing it to form a more efficient path. Therefore, all schemes achieve a shorter η . In particular, the performance gain of the non-cooperation scheme is greater than that of the other three schemes, as it eliminates the need for cooperation. Further, when $W_U \geq 25$ kg, the performance of the proposed scheme is the same as that of the non-cooperation scheme because the UAV can pick up all the parcels at once without dropping them in between. Fig. 4(d) shows that the η of all schemes increases as L increases. This is because the UAV with a limited load capacity needs to drop off parcels more frequently onto the UGV or at the destination. Consequently, the performance gap between the proposed scheme and existing schemes increases because of the proposed cooperation strategy. In addition, while the RRT* scheme generally performs second to the proposed scheme,

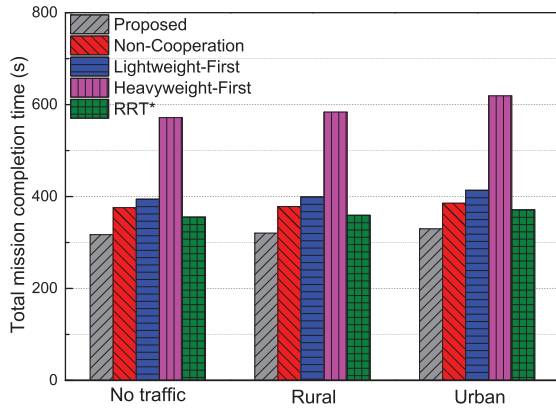


Fig. 5. Total mission completion time (η) versus traffic frequency.

it performs worse than the non-cooperation scheme when the UAV has relatively high degrees of freedom in path formation, e.g., when V_G is small and V_U is large, or when cooperation is relatively less important, e.g., when W_U is large and L is small. This is because the RRT* scheme requires a sufficiently large number of iterations to find the optimal path, but as the number of degrees of freedom in path optimization increases, 2×10^4 iterations are not enough to achieve this objective. However, it takes 52.1 seconds to achieve the result of the proposed scheme, while 2033.2 seconds to achieve the result of the RRT* scheme, showing the superiority of the proposed path planning in terms of optimality and computation time over the RRT* algorithm.

Fig. 5 shows the total mission completion time (η) versus traffic frequency, where $\lambda_0 = 1.28 \times 10^{-6}$ [traffic/m·s] for urban scenario and $\lambda_0 = 3.26 \times 10^{-7}$ [traffic/m·s] for rural scenario [32]. The shortest η is achieved when there is no traffic, and as the scenario moves from rural to urban to high-traffic environments, the η of all schemes increases due to delays caused by unexpected traffic. Finally, all results reveal that the proposed scheme outperforms the baseline schemes, which demonstrates the effectiveness of the proposed optimization methods.

V. CONCLUSION

In this paper, we explored the joint optimization of path planning and cooperative strategy for UAV-UGV delivery. We also considered practical scenarios, where the UAV can drop the collected parcels onto the UGV or at the destination to avoid being unable to pick up any more parcels owing to weight restrictions. Using the derived system model, we jointly optimized the horizontal and vertical trajectories of the two vehicles, along with binary indicators for pickup, drop-off, and cooperation, to minimize the total mission completion time. To address the nonconvexity of the formulated MINLP, we used the SCA method to transform the problem into a convex form and applied the PCCP technique to maintain the binary nature of integer variables. We also proposed a cooperative algorithm that iteratively finds a suboptimal solution using interior-point methods. Simulation results showed that the proposed scheme optimally determines the cooperative location for the UAV to drop off parcels onto the UGV by

jointly considering path planning and cooperation. This led to significant improvements in delivery time, with the proposed scheme outperforming the baseline schemes across various system parameters. An interesting topic for future work is to investigate cooperation strategies between multi-UAV and multi-UGV, taking into account realistic constraints such as energy consumption, weather/terrain variability, and unexpected obstacles. We anticipate that our study will contribute to the design of cooperative UAV-UGV systems, enabling more intelligent, secure, and efficient autonomous delivery solutions.

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